Siddarth Senthilkumar

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U.S. Citizen

EXPERIENCE

Google New York, NY
— Security Engineer *November 2023 - Present*

- Developing Automated Threat Modeling tooling that enabled the delivery of tens of threat models.
- Architecting a query evaluation system to detect non-compliance with security remediations.
- Authoring skills for security agents to review design docs, code, and other assets throughout the SDLC.
- Creating Terraform blueprints & architecture guides to document secure GCP practices.

Brex New York, NY
— Application Security Engineer *December 2021 - November 2023*

- Identified flaws in authorization, input validation, and other appsec concerns through design doc review.
- Pentested GraphQL APIs, browser extensions, OAuth flows, and vendor integrations.
- Designed & coded a worker-leader scanning infrastructure in Python to run Masscan, Nmap, & Nuclei.
- Communicated with 3rd party researchers to triage & ticket findings from bug bounty program.
- Developed and deployed vuln management and attack surface management tools.
- Wrote Semgrep rules to automatically catch security bugs during pull requests.

Microsoft Redmond, WA
— Software Engineer *June 2020 - December 2021*

- Designed attestation primitives with Intel, AMD, & OpenEnclave for Azure Confidential Compute.
- Built Azure service to cache Intel SGX collateral enabling Microsoft to control update timelines.
- Developed features for Azure host agent to securely refresh expired vTPM certificates.
- Designed protocol for unreliable one-way data transfer to Air-Gapped clouds with no internet access.
- Worked with Azure Attestation to balance timely security updates against inconveniences to customers.

Cruise Automation San Francisco, CA
— Application Security Intern *June 2019 - September 2019*

- Created a visualization tool to show impact of GCP IAM issues that allow privesc between projects.
- Created Docker images for vuln scanners to asynchronously alert the security team on issues.
- Performed security assessments; identified unauthed endpoints and faulty input validation.
- Introduced automated AuthZ testing for application endpoints to save time on future assessments.

Booz Allen Hamilton Annapolis Junction, MD
— Reverse Engineering Consultant *January 2018 - May 2018*

- Researched vulnerabilities in SCADA systems by analyzing firmware updates for the QNX RTOS.
- Reverse engineered FFS3 filesystem and wrote decoder to extract all files and directories from an image.
- Analyzed PowerPC ELFs to exploit weak client-side authentication routines.

EDUCATION

University of California, Santa Barbara Santa Barbara, CA
M.S. in Computer Science, GPA: 3.97 *September 2018 - June 2020*

Georgia Institute of Technology Atlanta, GA
B.S. in Computer Science, GPA: 3.84 *August 2014 - December 2017*

SKILLS

Languages	Python, C, C#, Java. Basics: TypeScript, Node.js, C++, HTML/CSS
Frameworks	Flask, React, Express, Material UI, ASP.NET MVC, OpenEnclave, JNI
Tools	Docker, Vagrant, Linux, GCP, AWS, Vim, Git, gdb, IDA Pro, Burp Suite